

Algebraic Topology: a beginner's course

This lecture:

**AlgTop34: More
homology computations**

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Lecture 34 More homology computations



$$\text{er } \partial_1 = \alpha(xy) + \beta(yz) + \gamma(zx) \\ + \delta(xw) + \varepsilon(wz) + \eta(wy)$$

$\downarrow \partial$

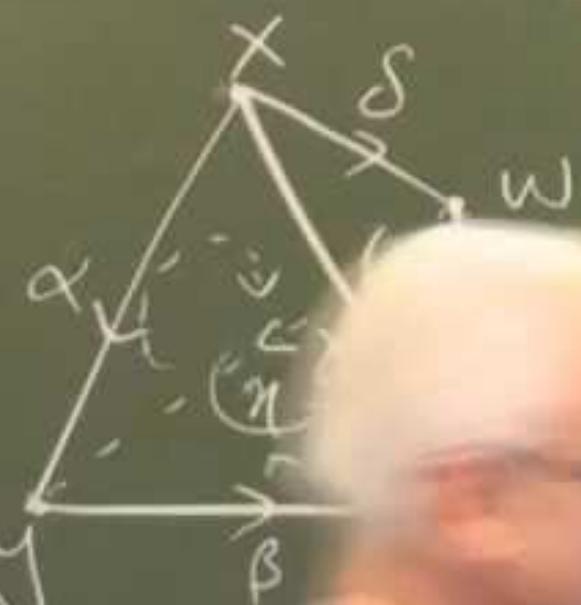
$$\alpha(y-x) + \beta(z-y) + \gamma(x-z) \\ + \delta(w-x) + \varepsilon(z-w) + \eta(y-w) = 0$$

$$x(-\alpha + \gamma - \delta) + y(\alpha - \beta + \eta) + z(\beta - \gamma + \varepsilon) \\ + w(\delta - \varepsilon - \eta) = 0$$

4 eq's in 6 unknowns

$$\begin{array}{c} \alpha \quad \beta \quad \gamma \quad \delta \quad \varepsilon \quad \eta \\ \left(\begin{array}{cccccc} -1 & 0 & 1 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & -1 \end{array} \right) \left(\begin{array}{cccccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \\ \downarrow \\ \left(\begin{array}{cccccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & -1 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & -1 \end{array} \right) \end{array}$$

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$$X = \{(xyz), (xzw), (y wz), (xwy)\}$$

$$C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0=0} 0$$

$$0 \rightarrow \mathbb{Z}^4 \xrightarrow{\partial_1} \mathbb{Z}^6 \xrightarrow{\partial_2} \mathbb{Z}^4 \rightarrow 0$$

$$\begin{aligned} &\langle (xyz), (xzw), (y wz), (xwy) \rangle \\ &\langle (xy), (yz), (zx), (xw), (wz), (wy) \rangle \end{aligned}$$

$$H_n = \mathbb{Z}_n \quad \mathbb{Z}_n = \ker \partial_n \text{ cycles}$$

$$B_n = \text{im } \partial_{n+1} \text{ boundaries}$$

$$H_0: \mathbb{Z}_0 = C_0 = \langle x, y, z, w \rangle$$

$$B_0 = \langle y, z, w \rangle$$

$$\mathbb{Z}_0 / B_0 = \mathbb{Z}$$

$$H_1 =$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \\ \epsilon \\ \eta \end{pmatrix}$$

homology computations



$$X = \{ (xyz), (xzw), (y wz), (xwy) \}$$

$$C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

$$0 \rightarrow \mathbb{Z}^6 \xrightarrow{\quad} \mathbb{Z}^4 \rightarrow 0$$

$\langle x, y, z, w \rangle$

$$\langle (xy), (yz), (zx), (xw), (wz), (wy) \rangle$$

$$Z_n = \ker \partial_n \text{ cycles}$$

$$B_n = \text{im } \partial_{n+1} \text{ boundaries}$$

$$H_1 = Z_1 / B_1, \quad Z_1 = \ker \partial_1$$

$$\begin{pmatrix} \alpha \\ \beta \\ \delta \\ \epsilon \\ \eta \end{pmatrix} = r \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

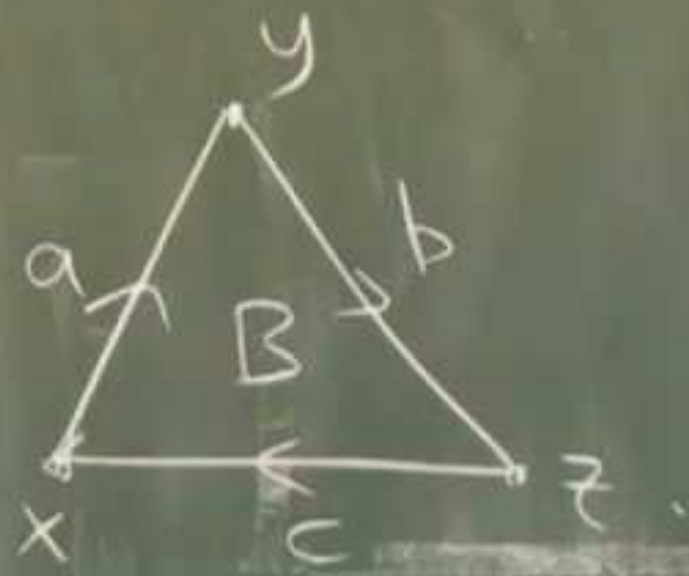
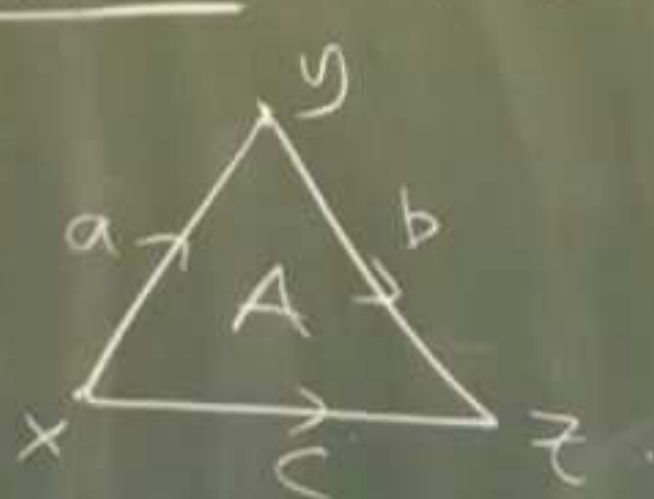
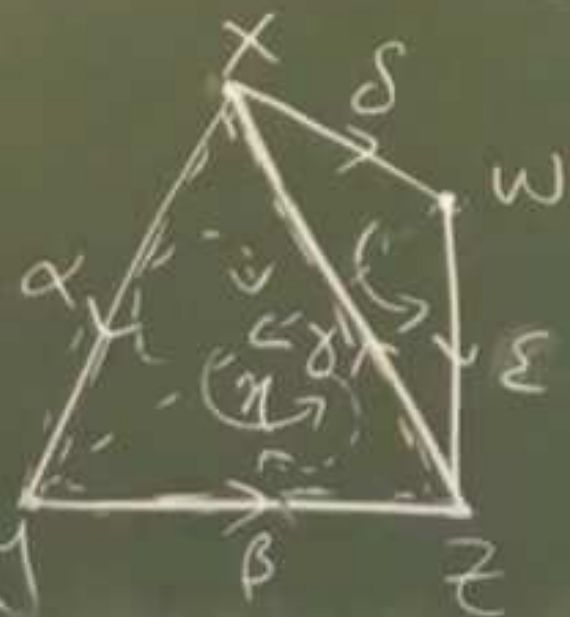
all solutions are multiples of cycles

$$\begin{aligned} & \alpha + \beta + \delta \\ & - \alpha - \beta + \delta + \epsilon \\ & - \alpha + \delta + \eta \\ \hline Z_1 &= \mathbb{Z}^3 \end{aligned}$$

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Eilenberg: Δ -complex:

Another $H_n(S^2)$:



$0 \rightarrow C_2 \rightarrow C_1 \rightarrow C_0 \rightarrow 0$

$C_2 \rightarrow C_1 \rightarrow C_0 \rightarrow 0$

$$H_2 = Z_2 / B_2$$

$$\partial((xyz) + (xzw) + (ywx) + (xwy))$$

$$Z_2 = \langle A+B+C+D \rangle \cong \mathbb{Z}$$

$$B_2 = 0 \quad \text{so } H_2(S^2) = \mathbb{Z}$$